

RegretGate

AI failures are found too late - after reply, send, refund, approve, or execute.

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github.com/harshalDharpure/controlplane-regretgate

1. Problem framing

Generative AI already moves money, data, and people inside enterprises through customer chat, employee copilots, and regulated decision-support. Failures often show up only after the model has already acted.

Oversight today is split across quality checks, cost dashboards, and safety filters. The gap: problems are detected after commit, when fixing them is costly and liability is real.

Three silent failure modes:

Performance

Confidently wrong
Hallucinations / poor decisions with certainty

Cost

Quietly expensive
Token burn, retries, tool thrash

Responsibility

Subtly unsafe
Bias, leakage, policy violations

Real-world complexities we design for

Different use cases

Different latency and risk budgets. One fixed checker does not fit all.

Overlapping risks

Bias, hallucination, and privacy often hit together, not as clean buckets.

No live ground truth

Many claims cannot be perfectly fact-checked in real time.

Over-flag vs under-flag

Too many flags cause bypass. Too few create liability. Must be tunable.

Agent compounding

One bad turn can shape several downstream decisions.

API-layer reality

Enterprises consume models via API, so we work on inputs/outputs and actions.

2. Solution - RegretGate

Expected Regret = P(failure) x Impact x Irreversibility (score 0-100)

RegretGate is a pre-commit control plane on pending actions (reply / send / refund / approve / execute / tool loop). It runs parallel checks for performance, cost/thrash, and responsibility, then routes through a regret-priced ladder.

Responsibility risks can hard-block on their own. Every decision leaves a receipt. HITL outcomes recalibrate thresholds so over/under-flagging can be tuned.

1. Intent

Identify pending action

2. Signals

P / C / R in parallel

3. Score

Expected Regret 0-100

4. Ladder

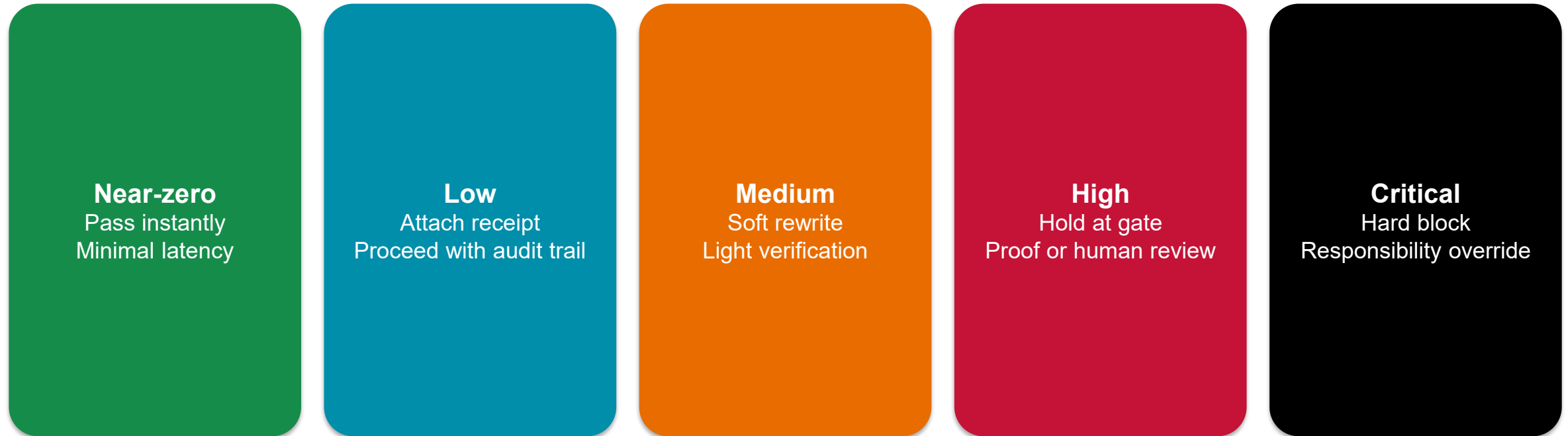
Pass to hard block

5. Audit

Receipt + HITL feedback

Intervention ladder

Verification effort follows Expected Regret. Fast where safe. Strict where irreversible.



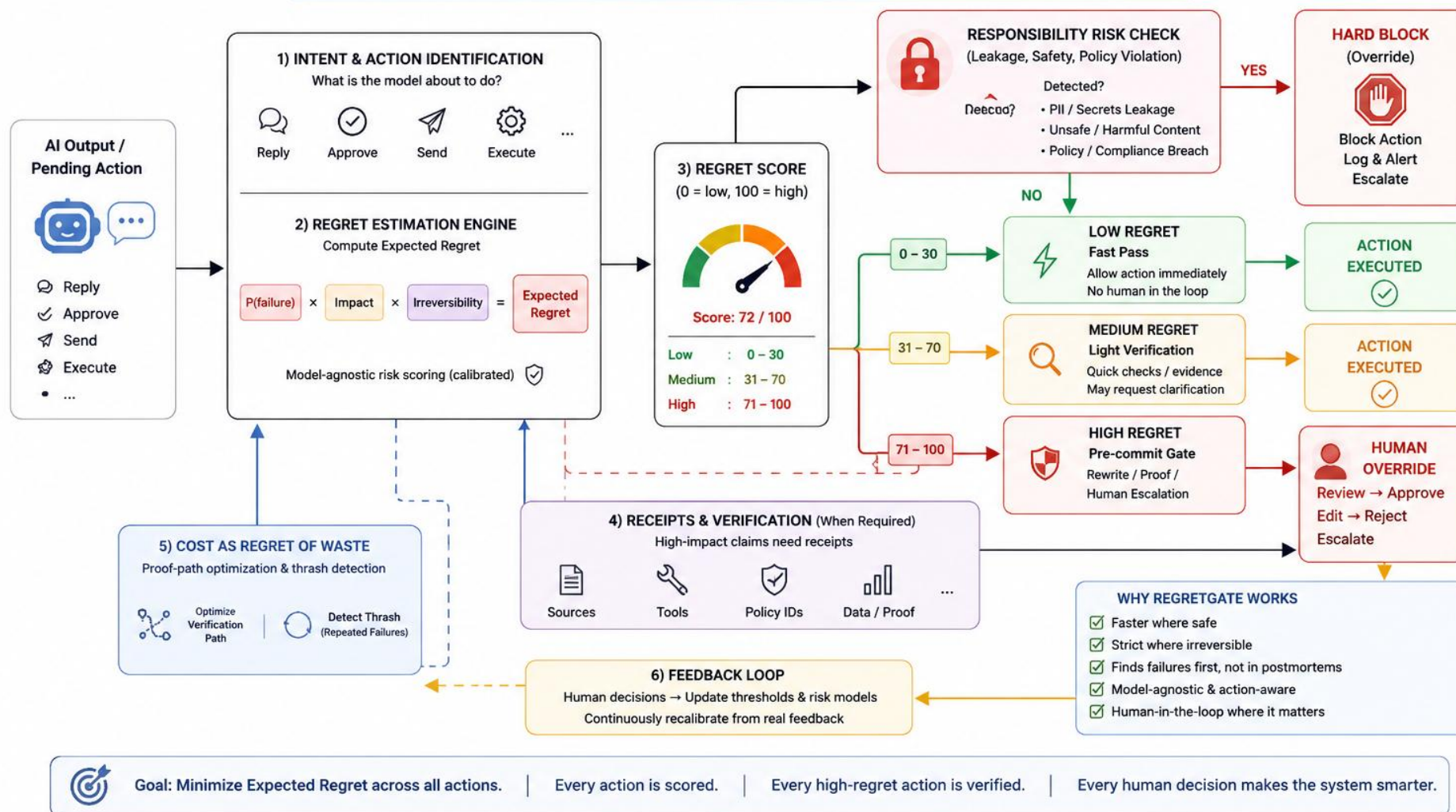
Same ladder as the prototype: score -> ladder -> hard block -> HITL -> feedback across support, copilot, and decision-support use cases.

Architecture

RegretGate (ControlPlane Checker)

Act with confidence. Verify only when the regret is high.

RegretGate reframes AI oversight around Expected Regret = $P(\text{failure}) \times \text{Impact} \times \text{Irreversibility}$.
Verify where it matters. Move fast where it's safe.



Pre-commit middleware: Intent -> parallel P/C/R signals -> Expected Regret -> ladder -> audit / HITL / feedback

Prototype results

Same ladder in the live app: Pass instantly -> Soft rewrite -> Hold at gate -> Hard block

Pass instantly

Safe FAQ | Regret 6

Pending AI action

Safe FAQ reply

Our returns window is 30 days for unused items. You can start a return from the Orders page.

EXPECTED REGRET

6/100

DECISION

Pass Instantly

Near-zero Expected Regret (6). Fast pass – verification effort deferred where safe.

Reply Customer Support Us Internal

40ms Path Allowed

NEAR-ZERO

LOW

MEDIUM

HIGH

CRITICAL

Pass instantly

Attach receipt

Soft rewrite

Hold at gate

Hard block

REGRET FACTORS

P(failure) 0.249

Impact 0.45

Irreversibility 0.25

Raw product 0.028

RESPONSIBILITY CHECK

No override

No PII / secrets / unsafe / policy breach

SIGNALS

No elevated tags

grounding 100% · waste 4%

USE CASE

Customer Support

POLICY PACK

US Internal

ACTION TYPE

reply

AMOUNT USD

0

TOKENS

120

TOOL CALLS

0

Soft rewrite

Confidently wrong balance | Regret 44

Pending AI action

Confidently wrong balance

Your account balance is definitely \$4,280.12 and the refund is guaranteed to post tonight according to our records.

EXPECTED REGRET

44/100

DECISION

Soft Rewrite

Medium Expected Regret (44). Soft rewrite applied for light verification before commit.

Reply Customer Support Us Internal

180ms Path Allowed

NEAR-ZERO

LOW

MEDIUM

HIGH

CRITICAL

Pass instantly

Attach receipt

Soft rewrite

Hold at gate

Hard block

REGRET FACTORS

P(failure) 0.782

Impact 0.8

Irreversibility 0.25

Raw product 0.1564

RESPONSIBILITY CHECK

No override

No PII / secrets / unsafe / policy breach

SIGNALS

Performance

Ungrounded

grounding 0% · waste 11%

USE CASE

Customer Support

POLICY PACK

US Internal

ACTION TYPE

reply

AMOUNT USD

4280

TOKENS

200

TOOL CALLS

0

Hold at gate

Tool thrash | Regret 65

Pending AI action

Quietly expensive tool thrash

Calling API search again. Retrying knowledge fetch. Trying again with the same query. Tool call: retrieve. Retrying once more.

EXPECTED REGRET

65/100

DECISION

Hold At Gate

High Expected Regret (65). Holding at pre-commit gate for proof or human escalation. Low grounding vs confident claims (no reliable real-time ground truth attached) Thrash detected: 3+ retries/tool calls with little new information

Tool_loop Internal Copilot Us Internal

800ms Path Blocked / Held

Human Review

NEAR-ZERO

LOW

MEDIUM

HIGH

CRITICAL

Pass instantly

Attach receipt

Soft rewrite

Hold at gate

Hard block

REGRET FACTORS

P(failure) 0.9623

Impact 0.55

Irreversibility 0.4

Raw product 0.2117

RESPONSIBILITY CHECK

No override

No PII / secrets / unsafe / policy breach

SIGNALS

Cost Thrash

Ungrounded

grounding 0% · waste 75% · thrash

USE CASE

Internal Copilot

POLICY PACK

US Internal

ACTION TYPE

tool_loop

AMOUNT USD

0

TOKENS

5200

TOOL CALLS

6

RETRIES

4

SOURCES

Grounding available

Hard block

PII send | Regret 78

Pending AI action

Subtly unsafe PII send

Hi Jordan, as requested we are emailing your SSN 312-55-9981 and card 4111 1111 1111 1111 for verification. Send email now.

EXPECTED REGRET

78/100

DECISION

Hard Block

HARD BLOCK (override): Potential PII detected in pending action output. Expected Regret score 78 is secondary to safety/policy.

Send Customer Support Us Internal

60ms Path Blocked / Held

Human Review

NEAR-ZERO

LOW

MEDIUM

HIGH

CRITICAL

Pass instantly

Attach receipt

Soft rewrite

Hold at gate

Hard block

REGRET FACTORS

P(failure) 0.9428

Impact 0.6

Irreversibility 0.5

Raw product 0.3111

RESPONSIBILITY CHECK

YES – hard block

Potential PII detected in pending action output

SIGNALS

Responsibility PII

Performance

Ungrounded

Overlap Hallucination Privacy

grounding 0% · waste 16%

USE CASE

Customer Support

POLICY PACK

US Internal

ACTION TYPE

send

AMOUNT USD

0

TOKENS

90

TOOL CALLS

1

RETRIES

0

SOURCES

Grounding available

3. Target users

AI Platform / MLOps

One consistent gate across apps without killing latency.

Risk / Compliance / Privacy

Audit trail, geo policy packs, hard blocks for leakage.

Business process owners

HITL only where irreversibility is high (refunds, approvals).

CIO / Responsible AI

Trust metrics and tunable over/under-flag tradeoffs.

4. Business case and impact

Latency preserved

Near-zero / low regret paths stay fast.

Liability reduced

High-regret approvals and refunds held before commit.

Cost waste cut

Thrash detection stops quietly expensive loops.

Alert fatigue managed

Effort scales with Expected Regret, thresholds tunable.

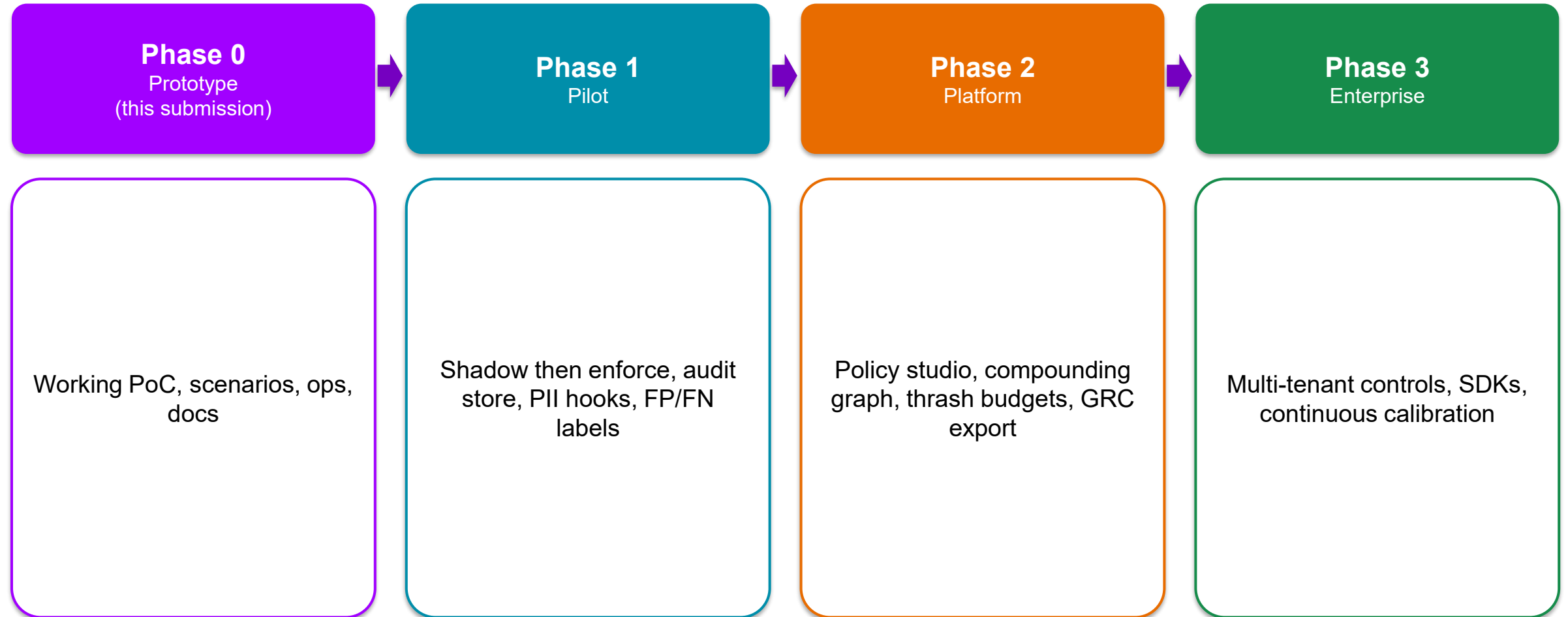
Illustrative framing (directional): 50,000 AI actions/week, ~3% high-regret, ~0.5% responsibility-critical.

Without a gate: incidents show up in postmortems (refund errors, leakage, agent spend spikes).

With RegretGate: high-regret cohort verified pre-commit; critical cohort hard-blocked; low-regret cohort unblocked.

Even a small drop in leakage or bad automated payouts usually beats checker operating cost.

5. Phased roadmap



6. Risks and mitigations

Over-flagging

Ladder + tunable thresholds + fatigue metrics

Under-flagging

Responsibility hard block independent of score

No ground truth

Receipts, soft rewrite hedges, HITL holds

Aging rules / drift

Feedback recalibration + versioned policy packs

Latency regression

Parallel checks and fast path for low regret

Gate bypass

Platform embedding + audit incompleteness alerts

Prototype and how to run

Round 2 demonstrates the core mechanism on sample data: score -> ladder -> hard block -> HITL -> feedback across three use cases and three silent failure modes. This is a prototype, not a claim of production ML accuracy.

Control Plane /

Paste or load a pending action. See score, ladder, rewrite, receipts.

Scenarios /scenarios

One-click demos: support, copilot, decision, overlap, policy packs.

Ops /ops

HITL queue, audit log, metrics, feedback, tighten/loosen policy.

GitHub: github.com/harshaIDharpure/controlplane-regretgate

Run: `npm install -> npm run dev -> http://localhost:3000`

No API keys required for the demo.

Thank you